# TAGADA: Transcript And Gene Assembly, Deconvolution, Analysis

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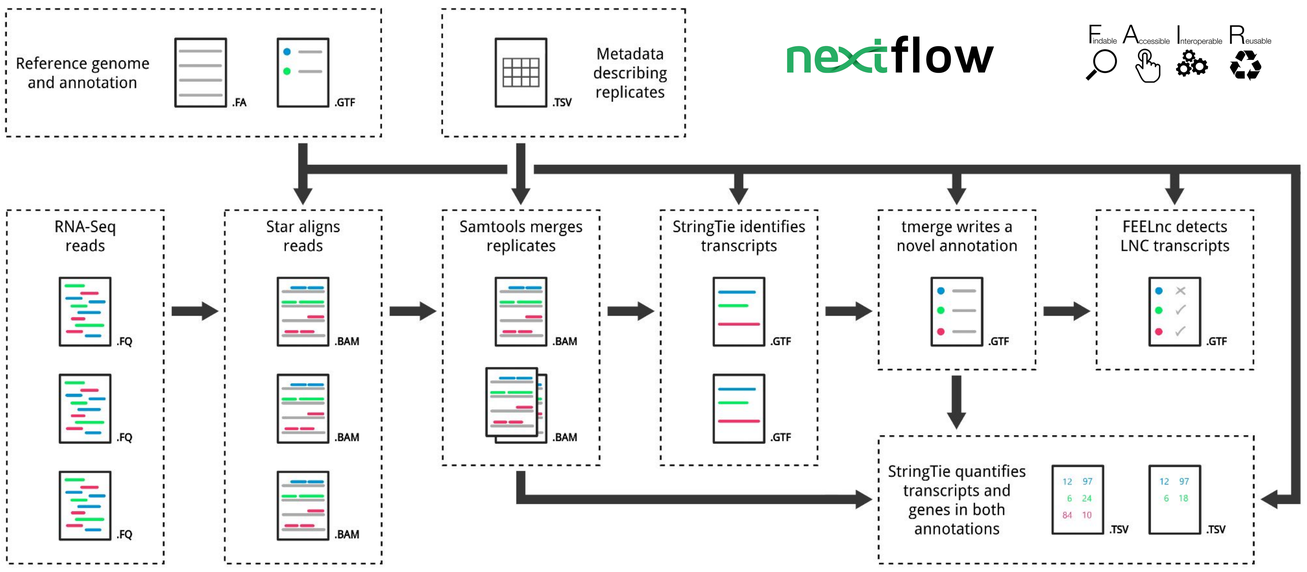
# An RNA-seq data analysis pipeline to improve genome annotations with new genes and transcripts

## Introducing the TAGADA pipeline

**T**ranscript **A**nd **G**ene **A**ssembly, **D**econvolution, **A**nalysis

- Integrates RNA-seq "short" reads into a reference annotation
- Identifies new genes and transcripts
- Predicts transcript coding/non-coding status
- Filters transcripts based on cross-replicates consistency
- Merges results from many samples into a single new annotation
- Quantifies gene expression on reference and new annotations
- Runs on various computational environments
- Open-source pipeline written in NextFlow

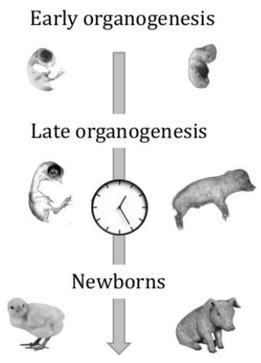



**Portable** **Reproducible** **Modular** **Scalable**

## Application to pig and chicken transcriptomes: the GENE-SWitCH project

The regulatory **GEN**om**E** of **SW**ine and **CH**icken

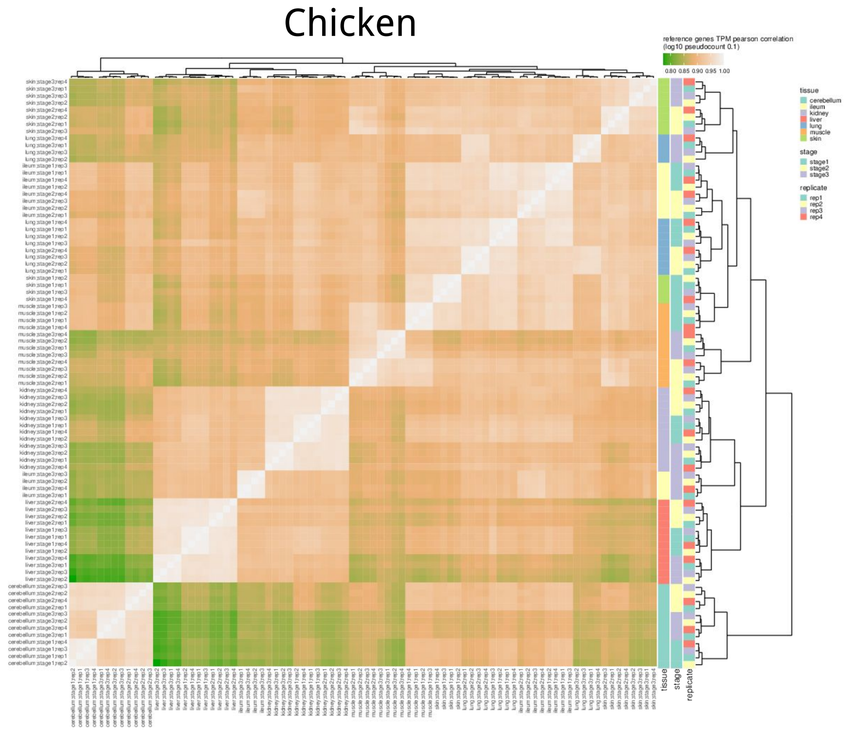
- Part of the FAANG international community
- 2 species: pig (*Sus scrofa*) and chicken (*Gallus gallus*)
- 4 animals per species: 2 males and 2 females
- 3 development stages: 2 embryonic and newborn
- 8 omics assays (including RNA-seq)
- 7 tissues



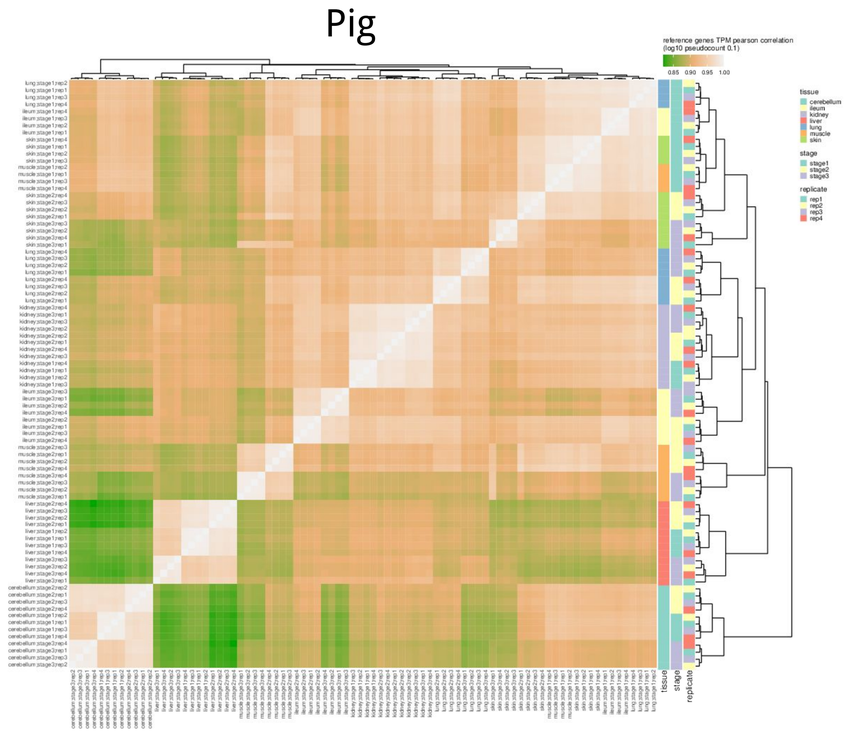

| | Pig | | Chicken | |
|---|---|---|---|---|
| | ENSEMBL | TAGADA | ENSEMBL | TAGADA |
| Genes | **31,908** | **51,171** | **24,356** | **34,712** |
| Expressed genes | **21,130** | **42,516** | **16,637** | **30,957** |
| Transcripts | **63,041** | **197,396** | **39,288** | **138,272** |
| Expressed transcripts | **40,396** | **171,103** | **24,973** | **124,874** |
| FEELnc coding transcripts | **-** | **3,152** | **-** | **3,636** |
| FEELnc non-coding transcripts | **-** | **50,105** | **-** | **40,197** |

Improving the reference annotation ENSEMBL v102 using TAGADA v2.1.0 with RNA-seq samples (~150M 2x150 bp x 84 libraries) on a Kubernetes cluster. New (non-)coding transcripts are predicted by FEELnc.

Chicken



Pig



Gene expression quantification: hierarchical clustering of samples from each species based on TPM expression values of reference genes. Figures are automatically generated by TAGADA during quality control.

## TAGADA in a nutshell

- is a containerized NextFlow pipeline that integrates RNA-seq data into an existing genome annotation
- tripled the number of annotated transcripts in pig and chicken by finding new transcripts of coding and non-coding genes
- generated expression profiles that led to meaningful sample hierarchical clustering
- is available at: https://github.com/FAANG/analysis-TAGADA

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